

---

Artificial Intelligence

---

BS (CS) Spring 2026

## Lab 05 Tasks



### Learning Objectives:

1. Informed Search
2. A\* Algorithm
3. Greedy Best-First Search

# Lab Tasks

## Submission Instructions

1. Create one notebook file for all questions with name *ROLLNO\_SEC\_LAB5* e.g. **i23XXXX\_A\_LAB05**
2. Submit your .ipynb file with correct name on Google Classroom under your section heading.
3. If you don't follow the above-mentioned submission instruction, you will be marked **zero**.

Plagiarism in the Lab Task will result in **zero** marks in the whole category.

## Questions

**Q1.** Design and implement a Python program to solve the 8-puzzle problem using the A\* search algorithm. The 8-puzzle consists of a 3×3 board containing eight numbered tiles (1–8) and one blank space represented by 0. A tile may move into the blank space if it is directly adjacent to it in one of the four directions: up, down, left, or right. The program should take the following initial state as input:

**Initial State**

|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
| 3 | 5 | 6 |
| 4 | 7 | 8 |

**Goal State**

|   |   |   |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 0 |

Based on problem analysis, select and implement an appropriate admissible heuristic function such as **Manhattan Distance** or **Euclidean Distance**. The Manhattan Distance calculates the sum of the absolute row and column differences between each tile's current position and its goal position, while the Euclidean Distance calculates the straight-line distance between the current and goal positions of each tile. The A\* evaluation function should be defined as  $f(n) = g(n) + h(n)$ , where  $g(n)$  represents the path cost from the initial state to the current state and  $h(n)$  represents the heuristic estimate to the goal.

The program should generate all valid successor states, apply the legal moves, and display the sequence of moves along with the total cost required to reach the goal state. Additionally, justify which heuristic is more suitable for the 8-puzzle problem and explain your choice based on efficiency and optimality considerations.

**Q2.** Given a set of cities and the distances between every pair of cities, implement a **Greedy Best-First Search** algorithm to find a route from a starting city to a destination city. The

heuristic values for all cities are provided in the table below. In this search, the evaluation function is defined as:

$$f(n) = h(n)$$

where  $h(n)$  is the heuristic value estimating the distance from city  $n$  to the goal city.

The problem requires finding a route from the starting city **Arad** to the destination city **Bucharest**, using the Greedy Best-First Search strategy. Your program should:

1. Represent the cities and their connections appropriately (e.g., as a graph).
2. Use the given heuristic values to guide the search.
3. Display the sequence of cities visited and the route selected by the algorithm.
4. Calculate the total distance traveled along the route.
5. Discuss whether Greedy Best-First Search guarantees the shortest path in this scenario.

#### Assumptions:

- All distances between cities are known.
- The heuristic values provided are an estimate of the straight-line distance to Bucharest.

